

MUONIO SAMMALTUNTURI, Finland

WMO: 028210

Lat: 67.973N

Lon: 24.116E

Elev: 1855

StdP: 13.74

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-4.5	-13.2	2.9	-8.6	-8.4	3.8	-3.2	43.8	15.1	37.5	18.8	17.4	80	0.972

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	72.1	58.2	68.0	56.6	64.3	54.7	60.7	68.7	58.3	65.3	56.3	62.2	11.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.0	77.1	62.8	55.4	69.9	59.6	53.5	65.3	57.9	27.7	69.1	26.0	65.7	24.7	62.4	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
34.8	31.2	28.0	DB	-11.5	74.3	5.6	3.8	-15.6	77.1	-18.9	79.3	-22.0	81.4	-26.1	84.2	
			WB	-12.0	62.8	5.3	3.5	-15.8	65.3	-18.9	67.4	-21.8	69.3	-25.7	71.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	12.3	13.2	18.9	28.2	38.9	47.3	54.7	50.8	42.6	29.7	21.9	17.3
	DBStd	16.25	8.34	9.52	6.71	6.51	8.85	7.91	7.48	6.51	5.80	6.46	6.42	7.46
	HDD50	7146	1170	1029	965	655	364	144	35	68	227	631	844	1014
	HDD65	12275	1635	1449	1430	1105	809	533	331	443	671	1096	1294	1479
	CDD50	361	0	0	0	0	21	62	181	91	6	0	0	0
	CDD65	16	0	0	0	0	1	1	12	2	0	0	0	0
	CDH74	44	0	0	0	0	2	3	34	5	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	15.5	17.4	17.5	17.5	15.1	14.8	14.4	13.4	13.0	14.2	14.8	16.0	17.9
Precipitation	PrecAvg	23.3	1.6	1.3	1.1	1.3	1.5	2.5	3.3	2.8	2.2	1.9	2.0	1.7
	PrecMax	29.8	3.1	3.0	2.4	3.0	3.8	4.2	5.6	5.7	4.4	3.7	3.8	3.0
	PrecMin	17.4	0.4	0.5	0.4	0.4	0.3	0.4	1.3	1.3	0.6	0.5	0.6	0.4
	PrecStd	3.6	0.6	0.6	0.6	0.6	0.8	0.9	1.2	1.1	1.1	0.8	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.3	35.6	35.6	51.7	71.0	73.3	78.0	72.8	58.0	45.1	39.1	33.7
		MCWB	30.8	34.3	31.3	41.5	51.5	58.5	60.7	61.0	50.8	42.9	38.1	32.8
	2%	DB	28.5	29.7	32.5	44.3	63.1	67.5	73.7	68.0	55.1	42.7	35.1	30.8
		MCWB	27.7	28.9	29.8	37.5	50.2	55.5	58.9	57.2	50.6	41.4	33.9	30.2
	5%	DB	26.1	27.7	30.2	40.2	57.6	63.3	70.0	63.7	52.8	40.9	32.0	28.8
		MCWB	25.5	27.2	28.5	35.0	47.3	52.8	57.5	55.5	48.9	39.8	31.6	28.4
	10%	DB	23.7	25.5	28.0	37.1	51.5	59.3	66.2	59.9	50.4	38.3	30.1	27.1
		MCWB	23.2	25.0	26.8	32.6	43.6	50.1	56.6	53.4	47.6	37.3	29.8	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	34.6	32.4	42.5	54.0	62.2	63.9	64.2	54.4	43.8	38.5	32.9
		MCDB	31.2	36.1	34.2	50.7	65.3	71.4	69.6	71.6	55.4	44.2	39.0	33.4
	2%	WB	27.9	29.1	30.5	38.4	50.8	57.8	61.4	59.2	51.8	41.7	34.0	30.3
		MCDB	28.3	29.6	32.1	43.3	61.6	64.7	69.3	64.4	53.7	42.5	34.7	30.7
	5%	WB	25.6	27.2	28.6	35.6	48.0	54.6	59.3	56.5	49.8	39.8	31.7	28.5
		MCDB	26.1	27.6	29.9	39.8	56.8	59.7	67.2	61.4	52.0	40.7	32.1	28.7
	10%	WB	23.2	25.0	26.9	33.2	44.3	51.7	57.6	54.6	48.2	37.4	29.8	26.8
		MCDB	23.8	25.5	27.9	36.4	50.6	58.5	65.0	58.7	49.9	38.2	30.1	27.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	7.6	8.2	9.9	11.7	12.3	12.1	10.6	7.8	6.0	6.9	6.9
		MCDBR	10.3	8.1	9.3	13.4	17.4	16.5	15.1	14.0	10.2	7.1	7.6	7.1
		MCWBR	10.5	8.0	7.7	8.2	9.4	8.8	7.2	7.0	6.6	5.9	7.6	7.0
	5% WB	MCDBR	10.3	7.9	8.2	12.8	16.1	14.7	13.4	12.5	8.1	6.7	7.7	7.1
		MCWBR	10.6	7.8	7.3	8.2	9.4	8.9	7.6	7.1	6.3	6.0	7.8	7.0
Clear-Sky Solar Irradiance	taub		0.176	0.234	0.270	0.299	0.317	0.321	0.330	0.315	0.287	0.254	N/A	N/A
	taud		1.917	2.106	2.213	2.244	2.447	2.515	2.500	2.521	2.572	2.338	N/A	N/A
	Ebn at Noon		40	183	246	270	279	281	274	266	248	180	N/A	0
	Edh at Noon		3	15	24	31	29	28	28	24	18	12	N/A	0
All-Sky Solar Radiation	RadAvg		12	159	635	1249	1551	1579	1477	1068	566	188	30	0
	RadStd		1	19	39	65	151	115	146	82	62	32	4	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page